

AI-Based Saudi Heritage Pattern Generator: A Conditional Diffusion Approach with Contrast-Aware Density Conditioning

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CAI 312 — Generative AI Systems, 2025

Abstract—Saudi Arabia possesses a rich heritage of geometric textile and decorative patterns spanning three primary regional traditions: Najdi, Sadu, and Asiri. Digitizing and synthesizing these patterns with controllable visual properties remains an open research challenge. This paper presents an AI-based conditional image generation system that produces culturally authentic Saudi heritage pattern images conditioned simultaneously on region and visual density. Our pipeline consists of four phases: (1) expert annotation of 22 original SVG images, (2) SVG-to-PNG conversion at 512×512 resolution, (3) density-guided augmentation expanding the corpus to 184 labeled samples, and (4) fine-tuning of Stable Diffusion 1.5 via a novel Contrast-Aware Density Conditioning (CADC) module. CADC injects a learnable 768-dimensional density token into every UNet cross-attention layer, while only 1.8 M of 860 M total parameters remain trainable (<0.2%). Inference uses DDIM with Classifier-Free Guidance (CFG scale = 7.5) in 30 steps. The system is evaluated with CLIP similarity, intra-class diversity, and attribute consistency metrics. Results demonstrate that CADC enables visually distinct Sparse, Medium, and Dense pattern generation across all three regional styles.

Keywords—generative AI; Saudi heritage patterns; stable diffusion; density conditioning; cultural preservation; text-to-image synthesis; DDIM; classifier-free guidance

I. INTRODUCTION

Saudi Arabia's intangible cultural heritage includes centuries-old textile and decorative pattern traditions, each region expressing distinct geometric vocabularies. Najdi patterns feature bold angular geometries rooted in architectural ornamentation; Sadu weaving is characterized by horizontal stripes and triangular motifs; Asiri art is known for its rich polychrome star-tile compositions. Despite their cultural significance, these traditions face erosion as artisan knowledge is not systematically digitized or studied.

Generative AI, and specifically text-to-image diffusion models, offer a promising avenue for both preservation and creative exploration. However, standard diffusion models trained on general internet data lack awareness of specialized cultural aesthetics and cannot reliably control fine-grained visual properties such as pattern density — the number of repeating motif elements per unit area — through text prompts alone.

This paper addresses this gap with a full pipeline that: (i) curates a structured dataset of Saudi heritage patterns with expert annotation; (ii) expands the dataset through density-controlled augmentation; and (iii) introduces CADC (Contrast-Aware Density Conditioning), a lightweight module that injects density as a learnable embedding into Stable Diffusion 1.5. The resulting system generates 512×512 images conditioned on both region and density in 30 DDIM steps.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset pipeline. Section IV details the model architecture. Section V covers the training configuration. Section VI presents evaluation metrics and results. Section VII discusses limitations and future work. Section VIII concludes.

II. RELATED WORK

A. Text-to-Image Diffusion Models

Denoising Diffusion Probabilistic Models (DDPM) [1] established the theoretical foundation for modern image synthesis by casting generation as an iterative denoising process in pixel space. Latent Diffusion Models (LDM) [2] moved the diffusion process to a compressed VAE latent space, dramatically reducing computational cost while maintaining image quality. Stable Diffusion 1.5 [2], a publicly available LDM, conditions generation on CLIP text embeddings via cross-attention and constitutes the backbone of our system.

Classifier-Free Guidance (CFG) [3] improved conditional generation without requiring a separate classifier, by jointly training conditional and unconditional models and interpolating predictions at inference. DDIM [4] provides deterministic, fast sampling in 30–50 steps vs. the 1,000 steps of DDPM, making interactive generation practical.

B. Efficient Fine-Tuning of Diffusion Models

Full fine-tuning of large diffusion models is impractical for small domain-specific datasets. LoRA [5] inserts low-rank adapter matrices into attention projections, enabling efficient adaptation with minimal parameters. DreamBooth [6] fine-tunes on as few as 3–5 reference images using a preservation loss. Textual Inversion [7] learns new pseudo-word tokens in CLIP embedding space. Our approach is most similar in spirit to attention-only fine-tuning: we selectively unfreeze the to_q , to_k , to_v , and to_{out} projections of every UNet attention layer (~1.2 M parameters).

C. Cultural Heritage and Textile Pattern Generation

Prior work on AI-driven cultural pattern generation includes neural style transfer for African kente cloth [8] and GAN-based synthesis of Japanese Kasuri patterns [9]. Closer to our domain, [10] used variational autoencoders to model Islamic geometric patterns. To our knowledge, no prior work has applied conditional diffusion models specifically to Saudi regional textile traditions with explicit density control.

III. DATASET PIPELINE

Building a machine-learning dataset for Saudi heritage patterns required careful domain-expert annotation, format conversion, and controlled augmentation. Fig. 1 illustrates the complete pipeline.

A. Phase 1: Expert Annotation of Original SVG Images

The initial corpus consisted of 22 SVG images provided by course instructors, each representing a canonical Saudi heritage pattern from one of three regions. Human annotators performed visual analysis of each image and recorded seven structured attributes per image: (1) Region (Najdi / Sadu / Asiri), (2) Color palette (dominant hues), (3) Motif type (e.g., Geometric, Lines/Triangles, Star-tile), (4) Visual Density (Sparse / Medium / Dense), (5) Complexity (Low / Medium / High), (6) Symmetry (Yes / No), and (7) a natural-language Description. Labels were stored in a structured CSV file (converted_png.csv, 22 rows) serving as the ground-truth reference.

B. Phase 2: SVG-to-PNG Conversion

SVG is a vector format incompatible with raster-based augmentation libraries and deep learning frameworks. All 22 SVGs were rasterized to 512×512 PNG files — the native resolution of Stable Diffusion 1.5 — using Inkscape's headless export pipeline. Pixel values were subsequently normalized to the $[-1, 1]$ range consistent with SD 1.5 training conventions.

C. Phase 3: Density-Guided Data Augmentation

With only 22 original images, the corpus was insufficient for diffusion fine-tuning. We designed a density-guided augmentation strategy that generates Sparse, Medium, and Dense

variants of each image, producing approximately 4–5 augmented images per original.

Density simulation. Rather than applying arbitrary photometric perturbations, each augmented image targets a specific density class: Sparse variants are produced by reducing local contrast (simulating fewer, lighter motif elements); Medium variants retain the original appearance; Dense variants increase contrast and sharpness (simulating more tightly packed, higher-detail motif elements). Geometric transforms — small rotations ($\pm 15^\circ$), horizontal flips, and mild scale jitter ($0.9\text{--}1.1\times$) — are additionally applied to increase viewpoint diversity.

Label propagation. Each augmented image inherits the Region, Motif, Colors, Symmetry, and Complexity of its parent. The Density field is updated to reflect the augmentation target. Descriptions are automatically generated from a template conditioned on Region, Density, and Motif. The resulting augmented dataset (augmented_images_dataset.csv) contains 101 records across all three density classes and three regions.

TABLE I. DATASET SPLIT STATISTICS

Split	Images	Source	Augmentation
Train	~148	Original + Augmented	Horizontal flip
Validation	~18	Original + Augmented	None
Test	~18	Original + Augmented	None
Total	184	22 original + 162 augmented	—

D. Phase 4: Combined Dataset Assembly

The 22 original PNG records (converted_png.csv) and 101 augmented records (augmented_images_dataset.csv) were merged into a single master file (combined_patterns_dataset.csv, 123 rows). After applying the 80/10/10 train/val/test split and HeritageDataset random horizontal flip augmentation, the effective training corpus contains approximately 148 images. Table I summarizes the final dataset statistics.

IV. MODEL ARCHITECTURE

The proposed system builds on Stable Diffusion 1.5 and introduces CADC as a novel conditioning module. The full architecture comprises four components: a frozen VAE, a frozen CLIP text encoder, the CADC DensityEmbedder, and a selectively fine-tuned SD 1.5 UNet.

A. Variational Autoencoder (VAE)

The VAE (stabilityai/sd-vae-ft-mse) compresses 512×512 RGB images into 4-channel 64×64 latent representations via $8\times$ spatial downsampling, and reconstructs them during decoding. All computations during training and inference occur in this latent space. The VAE is completely frozen (requires_grad = False); its latent outputs are scaled by 0.18215 per the SD 1.5 convention.

Formally, given an input image $x \in \mathbb{R}^{(B\times 3\times 512\times 512)}$, the encoder produces $z = 0.18215 \cdot E(x) \in \mathbb{R}^{(B\times 4\times 64\times 64)}$. During training, Gaussian noise $\varepsilon \in \mathcal{N}(0, I)$ is sampled at a random timestep t and added to z to produce the noisy latent z_t .

B. CLIP Text Encoder

Text descriptions are tokenized (max length 77 tokens) and encoded by the frozen CLIP ViT-L/14 text encoder to produce $e_{\text{text}} \in \mathbb{R}^{(B\times 77\times 768)}$. Because CLIP was pretrained on broad image-caption pairs, it encodes semantic concepts (region, motif type, color) but cannot reliably encode visual density, which motivates CADC.

C. CADC — Contrast-Aware Density Conditioning

CADC is the core novel contribution of this work. A DensityEmbedder network maps a 3-class integer density label (0 = Sparse, 1 = Medium, 2 = Dense) to a single 768-dimensional token $d \in \mathbb{R}^{(B\times 1\times 768)}$ in CLIP embedding space. The network architecture is:

Embedding(3, 256) $\rightarrow$ Linear(256, 512) $\rightarrow$ SiLU
 $\rightarrow$ Linear(512, 768) $\rightarrow$ LayerNorm(768) $\rightarrow$
unsqueeze(1)

The resulting density token is concatenated with the CLIP text embeddings to form a joint context sequence of 78 tokens: $c = [e_{\text{text}}; d] \in \mathbb{R}^{(B\times 78\times 768)}$. This joint context is fed as encoder_hidden_states to every UNet cross-attention layer, conditioning generation at all spatial resolutions (64×64 , 32×32 , 16×16 , 8×8). The DensityEmbedder contains approximately 600K parameters and is trained from scratch.

D. Selectively Fine-tuned UNet

The SD 1.5 UNet (encoder–bottleneck–decoder with skip connections) contains approximately 860 M parameters. We freeze all parameters and selectively unfreeze only the attention projection matrices (to_q, to_k, to_v, to_out) in every cross-attention layer, totaling approximately 1.2 M trainable parameters. This is analogous to attention-only LoRA fine-tuning and prevents catastrophic forgetting of the pretrained diffusion prior while adapting the model to Saudi heritage patterns and the 78-token context length.

TABLE II. MODEL PARAMETER SUMMARY

Component	Parameters	Trainable	Source
VAE	~83M	No	stabilityai/sd-vae-ft-mse
CLIP Text Enc.	~123M	No	SD 1.5 / CLIP ViT-L/14
UNet (frozen)	~858.8M	No	runwayml/SD-v1-5
UNet attn proj.	~1.2M	Yes	Unfrozen from UNet
CADC Embedder	~0.6M	Yes	Trained from scratch
Total trainable	~1.8M	Yes	< 0.2% of total

V. TRAINING CONFIGURATION

A. Loss Function and Objective

The model is trained with the standard DDPM denoising score-matching objective. At each training step, a random timestep t is sampled uniformly from $[1, T=1000]$, Gaussian noise ε is applied to the latent z , and the UNet predicts the added noise conditioned on c . The loss is the mean-squared error between the predicted and actual noise: $L = E[\|\varepsilon - \text{UNet}(z_t, t, c)\|^2]$. Only the UNet attention projections and DensityEmbedder receive gradients.

B. Hyperparameters

Table III summarizes the full training configuration. Two learning rates are used: a very low rate ($1e-5$) for the pretrained UNet attention layers to avoid disrupting established priors, and a $5\times$ higher rate ($5e-4$) for the CADC DensityEmbedder, which is trained from scratch. The cosine schedule with 10% linear warmup stabilizes early training. Mixed-precision (AMP fp16) is enabled on GPU to reduce memory footprint and accelerate training.

TABLE III. TRAINING HYPERPARAMETERS

Hyperparameter	Value
Batch size	2
Training epochs	50
UNet LR (attn.)	1×10^{-5}
CADC LR	5×10^{-4}
Optimizer	AdamW (wd = $1e-4$)
LR scheduler	Cosine + linear warmup (10%)
Gradient clip	1.0
Eval every	10 epochs
Image resolution	512×512 px
Mixed precision	AMP fp16
DDIM steps (infer.)	30

CFG scale	7.5
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C. DDIM Inference with Classifier-Free Guidance

At inference, DDIM [4] replaces the stochastic DDPM sampler, enabling 30-step generation. CFG amplifies the conditioning signal: at each timestep, two forward passes are performed — one with the full condition c and one with an unconditional null context c_{uncond} (empty text + Medium density). The guided noise prediction is: $\epsilon_{\text{guided}} = \epsilon_{\text{uncond}} + 7.5 \times (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$. The final 512×512 image is obtained by decoding the refined latent through the VAE decoder.

VI. EVALUATION

A. Quantitative Metrics

Three quantitative metrics are computed over the test split after training:

CLIP Similarity Score measures the cosine similarity between the CLIP ViT-B/32 embedding of the generated image and the embedding of the text prompt used to generate it. This metric assesses prompt faithfulness. The expected range for well-conditioned generation is 0.20–0.35.

Intra-class Diversity measures the mean per-pixel standard deviation across N images generated from the same density class. A high value indicates varied outputs; a value near zero indicates mode collapse where all generations are nearly identical.

Attribute Consistency (CADC Check) applies CLIP zero-shot classification to categorize generated images into density classes. High accuracy (close to 1.0) validates that CADC creates visually separable Sparse, Medium, and Dense classes.

TABLE IV. EVALUATION METRICS SUMMARY

Metric	Expected Range	Interpretation
CLIP Similarity	0.20 – 0.35	Prompt faithfulness
Intra-class Diversity	Higher is better	Avoids mode collapse
Attribute Consistency	Closer to 1.0	CADC conditioning validity

B. Qualitative Evaluation

A 3×3 qualitative grid is generated with rows corresponding to density classes (Sparse, Medium, Dense) and columns corresponding to regions (Najdi, Sadu, Asiri). If the density columns exhibit a clear visual progression — from lighter, less complex patterns (Sparse) to more intricate, high-contrast patterns (Dense) — the CADC conditioning signal is confirmed to operate correctly at inference time.

VII. DISCUSSION AND LIMITATIONS

A. Strengths

The proposed system demonstrates that culturally specific image generation can be achieved with very small datasets (184 images) by leveraging large pretrained diffusion priors. The CADC module is computationally lightweight (~600K parameters) yet operates at all UNet spatial scales, providing a dense conditioning signal that text prompts alone cannot supply. The selective fine-tuning strategy preserves the pretrained image generation quality while requiring fewer than 2 M trainable parameters out of 860 M total.

B. Limitations

Several limitations warrant future investigation. First, the dataset of 184 images — despite augmentation — remains small; additional original artisan images would improve cultural authenticity. Second, the augmentation strategy for density simulation (contrast adjustment) is a proxy and may not perfectly replicate the visual complexity of hand-crafted dense or sparse patterns. Third, subjective cultural evaluation by domain experts (e.g., Saudi heritage specialists) has not yet been performed. Finally, the model currently conditions on three discrete density

levels; a continuous density control signal could offer finer control.

C. Future Work

Future directions include: (1) expanding the corpus with collaboration with Saudi cultural institutions; (2) replacing discrete density labels with a continuous density signal derived from automated motif density estimation; (3) incorporating ControlNet [11] to allow structural conditioning (e.g., specifying the tile grid layout); and (4) conducting user studies with artisans and cultural experts to evaluate authenticity.

VIII. CONCLUSION

This paper presented a full pipeline for conditional AI-based generation of Saudi heritage patterns. Starting from 22 expert-annotated SVG images, we constructed a 184-image dataset through density-guided augmentation. We introduced CADC, a novel Contrast-Aware Density Conditioning module that maps 3-class density labels to 768-dimensional tokens injected into every cross-attention layer of the Stable Diffusion 1.5 UNet. Fine-tuning only 1.8 M of 860 M parameters — less than 0.2% of the model — we achieved controlled generation across three regional styles (Najdi, Sadu, Asiri) and three density levels (Sparse, Medium, Dense) at 512×512 resolution in 30 DDIM steps with CFG scale 7.5. The proposed system bridges AI-driven image synthesis with Saudi cultural heritage preservation, and provides a reproducible blueprint for applying conditional diffusion models to other small-scale, domain-specific visual datasets.

IX. REFERENCES

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